

Chapter 82

Using the Tracker Node

This chapter shows the many capabilities of the Tracker node in Fusion, starting with how trackers can be connected in your node trees, and finishing with the different tasks that can be performed.

Contents

Introduction to Tracking	1725
Tracker Node Overview	1726
Modes of the Tracker Node	1726
Basic Tracker Node Operation	1727
Connect to a Tracker's Background Input.....	1727
Analyze the Image to be Tracked	1727
Apply the Tracking Data	1728
Viewing Tracking Data in the Spline Editor	1731
Tracker Inspector Controls	1732
Motion Tracking Workflow In Depth	1733
Connect the Image to Track	1733
Add Trackers	1733
Refine the Search Area	1736
Perform the Track Analysis	1736
Tips for Choosing a Good Pattern	1737
Using the Pattern Flipbooks	1738
Using Adaptive Pattern Tracking.....	1739
Dealing with Obscured Patterns	1740
Dealing with Patterns That Leave the Frame	1740
Setting Up Tracker Offsets.....	1741
Stabilizing with the Tracker Node	1742
Stabilization Using the Tracker Match Move Mode.....	1742

Smoothing Motion	1743
Using the Tracker Node for Match Moving	1744
Simple Match Moving	1744
Corner Positioning Operations	1745
Perspective Positioning Operations	1745
Connecting to Trackers' Operations	1745
Steady Position	1745
Steady Angle	1745
Offset Position	1746
Unsteady Position	1746
Steady Size	1746
Using the Outputs of a Tracker	1746
Using the Tracker as a Modifier	1748
Match Moving Text Example	1751
Adding a Layer to Match Move	1751
Setting Up Motion Tracking	1751
A Simple Tracking Workflow	1753
Connecting Motion Track Data to Match Move	1756
Offsetting the Position of a Match Moved Image	1758

Introduction to Tracking

Tracking is one of the most useful and essential techniques available to a compositor. It can be roughly defined as the creation of a motion path from analyzing a specific area in a clip over time. Fusion includes a variety of different tracking nodes that let you analyze different kinds of motion. Once you have tracked motion on a clip, you can then use the resulting data for stabilization, motion smoothing, matching the motion of one object to that of another, and a host of other essential tasks.

Types of tracking nodes in Fusion:

Tracker: Follows a relatively small, identifiable feature or pattern in a clip to derive a 2D motion path. This is sometimes referred to as point tracking.

Planar Tracker: Follows a flat, unvarying surface area in a clip to derive a 2 ½D motion path including perspective. A planar tracker is also more tolerant than a point tracker when some tracked pixels move offscreen or become obscured.

Camera Tracker: Tracks multiple points or patterns in a clip and performs a more sophisticated analysis by comparing those moving patterns. The result is a precise recreation of the live-action camera in virtual 3D space.

Each tracker type has its own chapter in this manual. This chapter covers the tracking techniques with the Tracker node.

Tracker Node Overview

The Tracker node is a single node that actually performs tracking, stabilizing, matching moving, and corner-pinning operations. Since the Tracker node can transform the foreground input, it can be used to generate tracks and then operate as a Merge in a match move or corner-pin setup. Or you can use it to produce tracking data only and then publish that data to other nodes in the Node Editor.

Modes of the Tracker Node

The Tracker node is an incredibly flexible tool often used multiple times in a composite to help with dozens of tasks. However, most of those tasks can be boiled down into just a few operations. The Tracker node has four operation modes that cover the majority of tracking situations.

Stabilizing

You can use one or more tracked patterns to remove all the motion from the sequence or to smooth out vibration and shakiness. When you use a single tracker pattern to stabilize, you stabilize only the X and Y position. Using multiple patterns together, you are able to stabilize position, rotation, and scaling.

Match Moving

The reverse of stabilizing is match moving, which detects position, rotation, and scaling in a clip using one or more patterns. Instead of removing that motion, it is applied to another image so that the two images can be composited together.

Corner Positioning

Corner positioning tracks four patterns that are then used to map the four corners of a new foreground into the background. This technique is generally used to replace signs or mobile phone screens. The Planar Tracker node is often a better first choice for these types of tracking tasks.

Perspective Positioning

Perspective positioning again tracks four patterns to identify the four corners of a rectangle. Each corner is then mapped to a corner of the image, rescaling and warping the image to remove all apparent perspective. The Planar Tracker node is often a better first choice for removing perspective from a clip.

Basic Tracker Node Operation

All tracking workflows consist of three fundamental steps.

- 1 Attach an image you want to track to the yellow background input of the Tracker node.
- 2 Set the tracking pattern and analyze the clip to create a path.
- 3 Apply the tracking data to stabilize, match move, corner pin, or remove perspective.

Connect to a Tracker's Background Input

You start by connecting the output of the image you want to track to a Tracker node's background input. The Tracker node analyzes the image that's attached to its background input.

You can insert the Tracker node serially with other nodes if you intend to use the Tracker node itself to do a simple stabilization operation or if you want to use it to perform the function of a Merge node in a match move or corner-pin operation.



Tracker node connected serially so it can both track and transform the input images

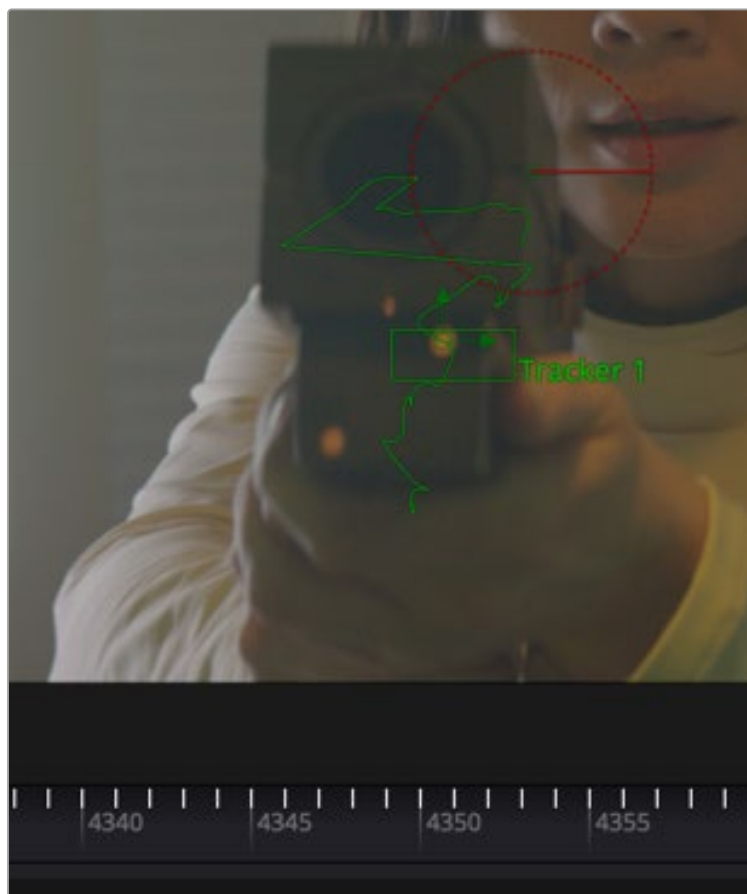
However, if you're just using a Tracker node to analyze data for use with multiple nodes elsewhere in the comp, you could choose to branch it and leave its output disconnected to indicate that Tracker node is a data repository. Please note that this is not necessary; serially connected Tracker nodes can be linked to multiple other nodes as well.



Tracker connected as a branch to indicate it is linked to other nodes and not used directly

Analyze the Image to be Tracked

After constructing the node tree and inserting the Tracker where you want, you can set up the tracker in the viewer. You identify one or more features in the image that you wish to track (referred to as patterns) by adding trackers (there's one by default) and positioning them using the onscreen controls in the viewer. After the Tracker node analyzes the clip, the resulting tracking data is stored within that Tracker node. Keyframes, one per frame, indicate the Tracked Center X and Y data that has been saved, while a motion path shows the path of tracked data in the viewer.



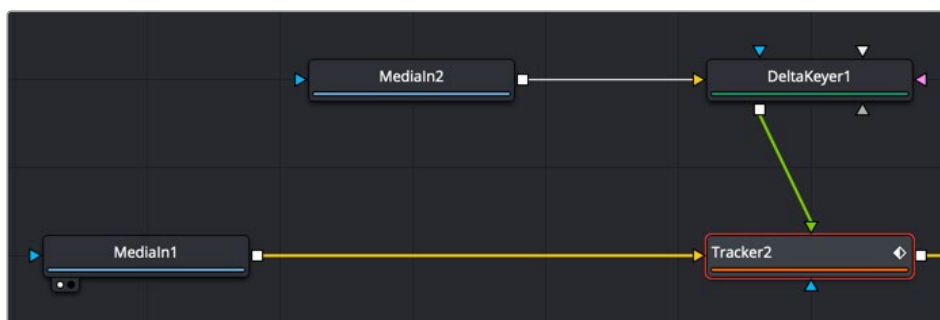
A motion path that indicates the tracked motion, and tick marks that indicate tracking data keyframes

Apply the Tracking Data

The resulting tracking data stored within the Tracker node is used to stabilize, match move, corner pin, or remove perspective in one of two ways.

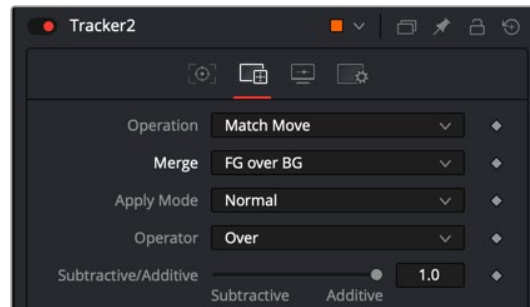
Method 1: Use the Tracker node to match move and merge

You can connect a foreground image to the Tracker node and apply the motion from the analyzed background image.



Using a Tracker node in line for a match move

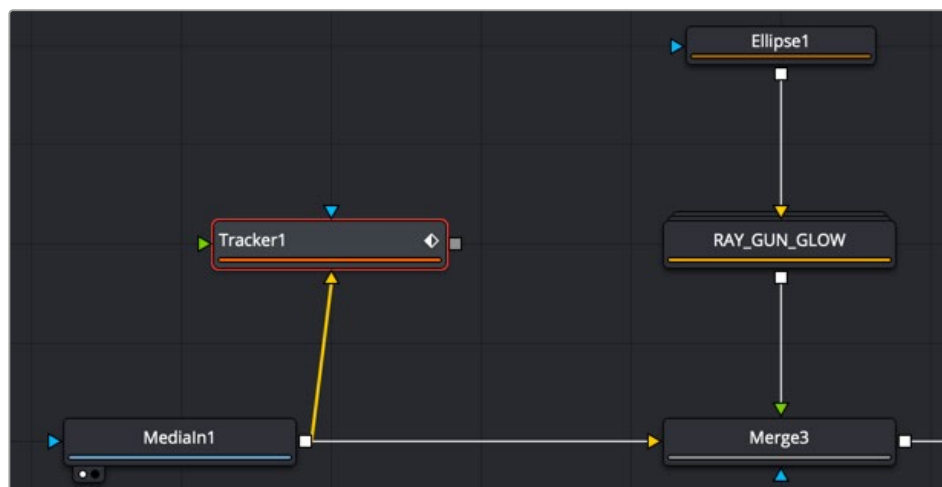
Setting the Operation parameter in the Operation tab in the Inspector to Match Move, Corner Position, or Perspective Position always applies the motion to the foreground input (if one is connected). This is an easy workflow for simple situations. In this scenario, you can use the Tracker node to replace a Merge node since Tracker nodes include all the same functionality as a Merge.



Using a Tracker node to do a match move and merge, all in one

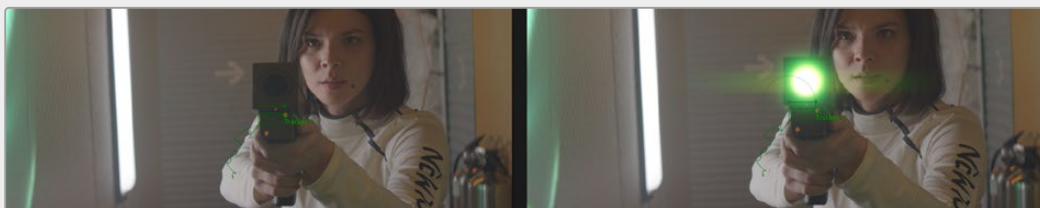
Method 2: Connect specific parameters to the Tracker Node

Alternatively, you can connect the tracking data from the Tracker node to the specific parameters of other nodes that will actually do the work, for instances where setting up a match move isn't just a matter of transforming a foreground image. Each Tracker node and each pattern within the Tracker node publishes its data for other nodes to use without directly linking to them in the node tree. For example, in the following node tree, an Ellipse node is being used to isolate a glow effect for the ray gun prop.



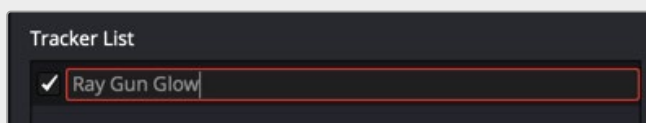
The Tracker set up as a branch and connected using the Connect To menu

The ellipse needs to follow the motion of the ray gun, so a Tracker node is used to analyze the movement of the gun tip so that tracking data can be used to animate the ellipse. The ellipse is not connected to the tracker directly via the foreground input but indirectly through the Connect To contextual menu.



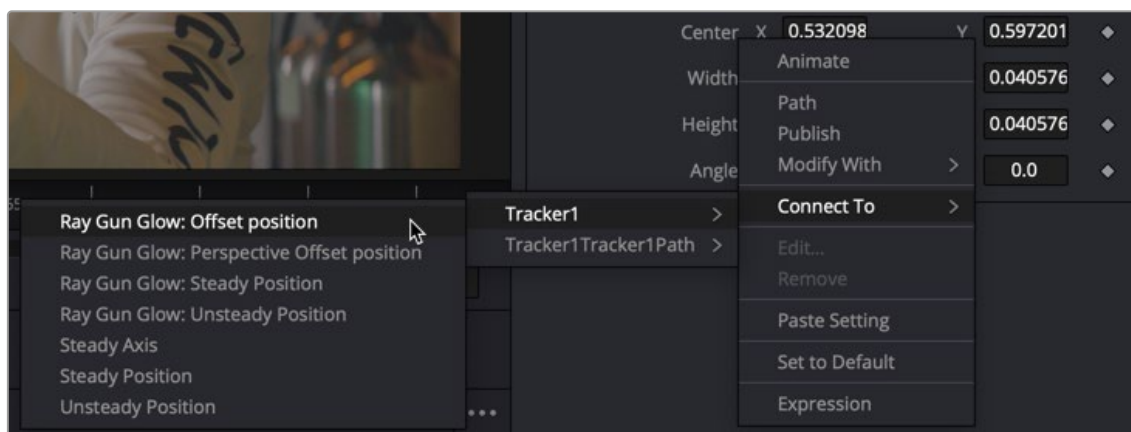
Applying the light of a ray gun by connecting tracking data to the center position of an Ellipse node

This is made easier by renaming the Tracker you created to something descriptive of what's being tracked.



You can rename trackers in the Tracker List by double-clicking them and typing descriptive text

Once the tip of the ray gun has been tracked, this tracking data is then connected to the Center parameter of an Ellipse node that's limiting a Glow effect by right-clicking the label of the Center parameter in the Inspector, and choosing Tracker1 > Ray Gun Glow: Offset position from the Connect to submenu of the contextual menu. All the data from every Tracker node in your node tree and every tracking pattern appears within this submenu, and since we named the Tracker, it's easy to find. Choosing Offset position because it will place the center of the ellipse directly over the path. However, it also gives us the flexibility to offset the ellipse if need be, using the offset controls in the Inspector.



Right-clicking a parameter's label lets you connect tracking data to it to animate it

You can connect the data from a Tracker node to any other node's parameter; however, you'll most typically connect track data to center, pivot, or corner X/Y style parameters. When you use tracking data this way, it's not necessary to connect the output of the Tracker node itself to anything else in your node tree; the data is passed from the Tracker to the Center parameter by linking it with the Connect To submenu.

Viewing Tracking Data in the Spline Editor

Tracking data can be seen in the viewer as a path or as a displacement spline in the Spline Editor. You can manipulate the tracking data in either place.

The Tracker uses a displacement spline by default that indicates how far the tracking point is based on the original location. It is great for modifying velocity, but it doesn't tell you anything about direction. If you need to nudge a few points in a certain direction, you can convert the displacement spline to an X and Y coordinate spline.



Right-clicking a parameter's label lets you connect tracking data to it to animate it

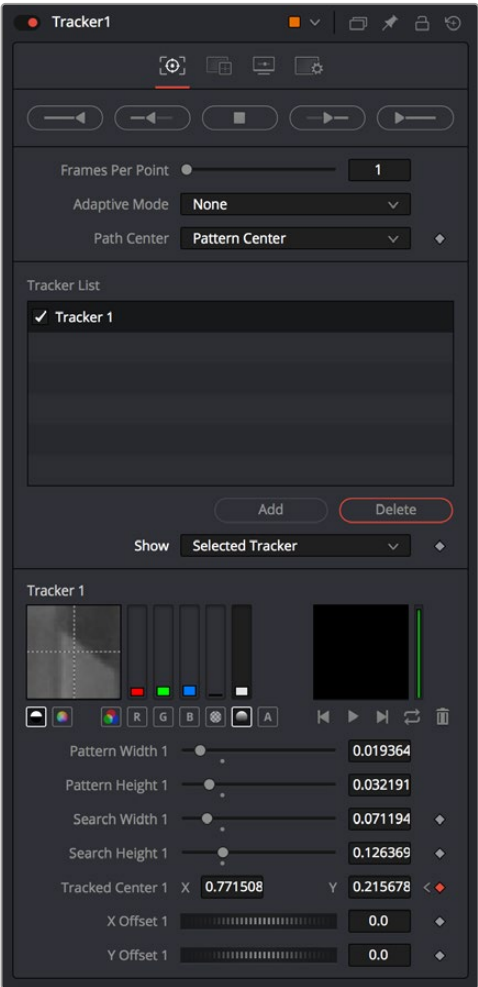
Right-click in the viewer to bring up a contextual menu. At the very bottom is a reference to the path the Tracker created, called Tracker1Tracker1Path:Polyline. Choosing it calls up a longer submenu where you can choose Convert to XY Path.

For more information on Displacement Splines, see *Chapter 72, "Animating in Fusion's Spline Editor,"* in the DaVinci Resolve Reference Manual or Chapter 10 in the Fusion Reference Manual.

Tracker Inspector Controls

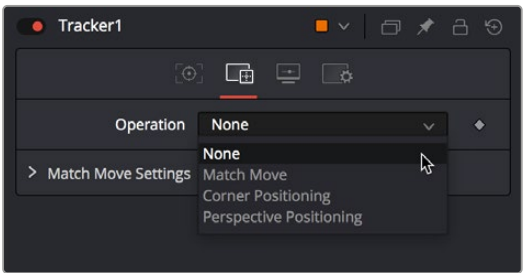
The layout of the Tracker node's tabs in the Inspector reflects this workflow. It's divided into three main Tracker tabs, as well as the common Settings tab.

- **The Tracker Control tab:** This is where you create onscreen trackers with which to target patterns, and where the controls appear that let you perform the required track analysis.

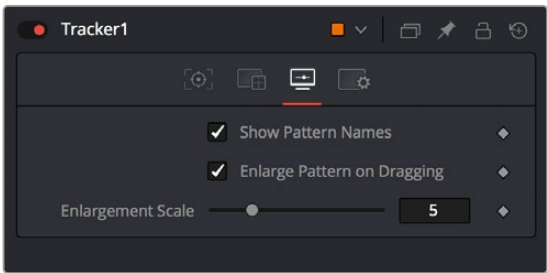


The Tracker Control tab

- **The Operations tab:** This is where you decide how the tracking data is used.



The Tracker Operations tab



The Tracker Display Options tab

- **The Display Options tab:** This is where you can customize how the onscreen controls look in the viewer.

Motion Tracking Workflow In Depth

Tracker nodes serve two purposes. They provide a method to analyze an object you want to follow, and they serve as a container for the resulting track data. This allows you to use one node for analysis and to pass on that analysis to any other node that requires it. Following is a more detailed breakdown of the tracking process.

Connect the Image to Track

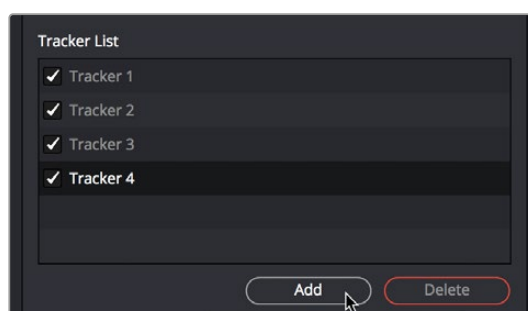
Regardless of whether you actually use the Tracker node itself to do anything with the tracking data, the image you want to track must be connected to the background input (yellow) of a Tracker node for there to be a successful analysis. While the Tracker node has a foreground input, it is initially ignored for purposes of tracking analysis, so even if you connect a foreground, the background is the only input used during the analysis process.



Connecting a MediaIn node's output to a Tracker node's background input

Add Trackers

Although each Tracker node starts with a single tracker pattern, a single node is capable of analyzing multiple tracking patterns that have been added to the Tracker List, enabling you to track multiple features of an image all at once for later use and to enable different kinds of transforms. Additional trackers can be added by clicking the Add button immediately above the Tracker List control.



To add an additional tracker, click the Add button in the Inspector

Multiple patterns are useful when stabilizing, match moving, or removing perspective from a clip. They also help to keep the Node Editor from becoming cluttered by collecting into a single node what would otherwise require several nodes.

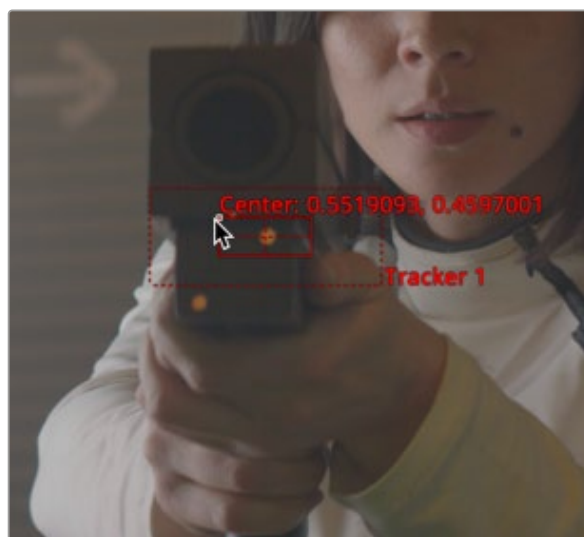
Working in the Tracker list:

- **To select a tracker:** Click the name of the Tracker you want to select.
- **To rename a tracker:** You can rename trackers to make it easier to reference them later. For example, if you're tracking a car door handle, you can name the Tracker "Car Handle" so it's easy to find later. To do so, just double-click the default name of the Tracker in the Tracker list, type a new one, and press Return.
- **To delete a tracker:** Select a tracker and click the Delete button.
- **To disable, suspend, or re-enable trackers:** Click the checkbox to the left of each tracker's name in the Tracker List. It's a three-way toggle that switches between Enabled, Suspended, and Disabled.
- **Enabled:** An Enabled tracker will re-track its pattern every time the track is performed. Its path data is available for use by other nodes, and the data is available for stabilization and corner positioning.
- **Suspended:** When the checkbox appears gray, it is Suspended. In this state, it does not re-track its pattern when the track is performed. The data is locked to prevent additional changes, but the data from the path is still available for other nodes. The data is also available for advanced tracking modes like stabilization and corner positioning.
- **Disabled:** A Disabled tracker does not create a path when tracking is performed. Its data is not available to other nodes or for advanced tracking operations like stabilization and corner positioning.

Position the Pattern Boxes

A pattern is the region of pixels that are targeted for tracking within an image using a pattern box. The pattern box is defined in the viewer by a rectangle when a tracker is active. A single Tracker node can have many pattern boxes, each targeting different patterns. Each tracked pattern will produce its own motion path.

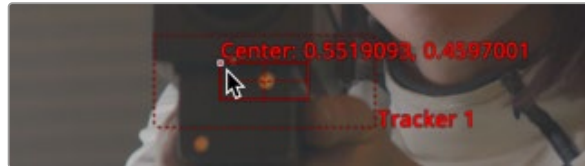
Clicking any part of a tracker's onscreen controls will select it. Selected pattern boxes are red, while deselected pattern boxes are green.



A pattern box positioned over an eye you want to track

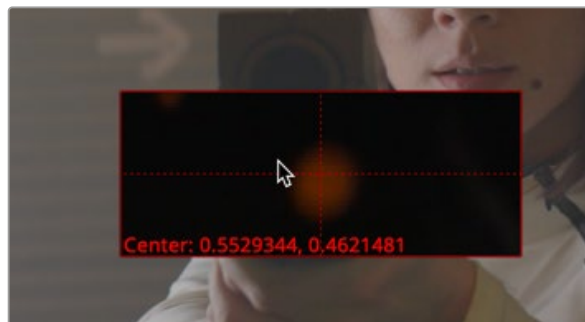
When you add a Tracker node to the Node Editor, you start with one pattern box displayed in the viewer as a small rectangle. When the cursor is placed over the pattern rectangle, the control expands and two rectangles appear. The outer rectangle has a dashed line, and the inner rectangle has a solid line. The outer rectangle is the search area, and the inner rectangle is the pattern.

If you need to select a new pattern, you can move the pattern box by dragging the small (and easily missed) handle at the top left of the inner pattern box.



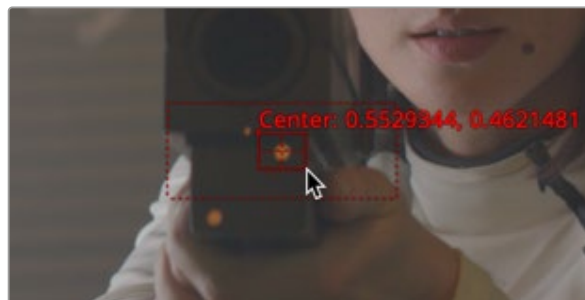
To move a pattern in the viewer, drag it from the upper-left corner

While moving the pattern box, an overlay pop-up appears, showing a zoomed version of the pixels contained within the rectangle to help you precisely position the pattern via the crosshairs within.



A zoomed view appears while you drag a pattern box to help you position it

The pattern rectangle can also be resized by dragging on the edges of the rectangle. You want to size the pattern box so that it fits the detail you want to track, and excludes area that doesn't matter. Ideally, you want to make sure that every pixel of the pattern you're tracking is on the same plane, and that no part of the pattern is actually an occluding edge that's in front of what you're really tracking. When you resize the pattern box, it resizes from the center, so one drag lets you create any rectangle you need.



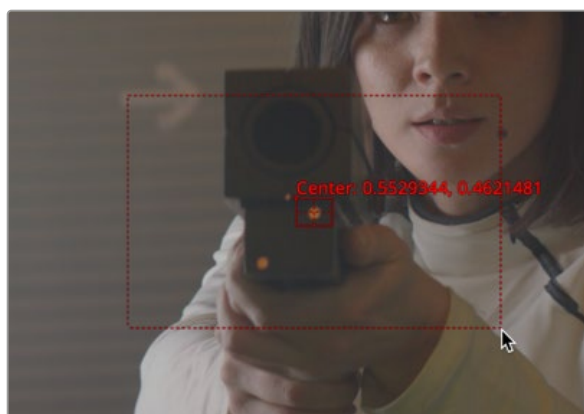
Resizing a pattern box to fit the tracking point on the ray gun

TIP: The magnified pattern box does not take viewer LUTs into account. When using Log content, it may make it easier to position the tracker if you temporarily insert a Brightness Contrast node between the source content and the yellow input of the tracker. You can use the Brightness Contrast node to temporarily increase the visibility of the region you are tracking.

Refine the Search Area

A second rectangle with a dotted border surrounds the pattern box. This is the search area. When progressing from one frame to another while tracking, the Tracker analyzes the region defined by the search area, which surrounds the last known tracker position in an attempt to relocate the pattern. The larger the search area, the better chance you have of successfully tracking fast moving objects, but the longer the track will take. However, there are some ways to optimize tracking for specific content.

For example, tracking a pattern that is moving quickly across the screen from left to right requires a wide search area but does not require a very tall one, since all movement is horizontal. If the search area is smaller than the movement of the pattern from one frame to the next, the Tracker will likely fail and start tracking the wrong pixels, so it's important to take the speed and direction of the motion into consideration when setting the search area.

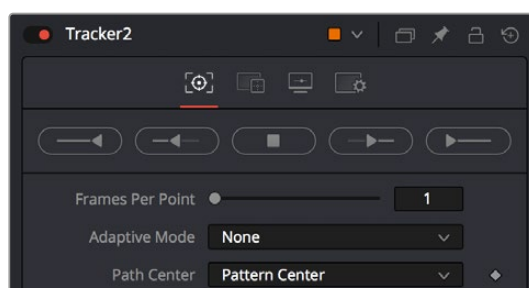


You can resize the search area by dragging the edges of the dotted outline

Perform the Track Analysis

Before you begin analyzing, you'll need to make sure you've set a render range in the Time Ruler that corresponds to the range of frames during which the pattern is visible. This may be an entire clip or only a small portion of that clip. Depending on the type of motion you're tracking, you may want to use the Adaptive Mode option to aid the analysis (see below for more details).

Once your options are set, you can use any of the tracking transport buttons at the top of the Inspector to start tracking. Once tracking has started, you cannot work in the Node Editor until it has completed.



The tracking transport buttons and analysis parameters

To begin tracking, do one of the following:

- Click the Track Reverse button to track from the very end of the render range.
- Click the Track Backward from Current Frame button to track backward from the current playhead position.
- Click the Track Forward button to track from the very start of the render range.
- Click the Track Forward from Current Frame button to track forward from the current playhead position.

Pattern tracking will stop automatically when it reaches the end of the render range (or the start when tracking backward), but you can also interrupt it and stop tracking at any time.

To stop tracking, do one of the following:

- Click the Stop Tracking button in the tracker transports.
- Click Stop Render at the bottom of the Fusion window.
- Press the Escape key.

When tracking is complete, the path will be connected to the pattern. The path from that pattern can now be connected to another node or used for more advanced operations like stabilization and corner positioning.

Once the track is complete, assuming it's good, you can use the various techniques in this chapter to use the track in your composition.

Tips for Choosing a Good Pattern

The Tracker works by searching each frame for the pixels contained in the pattern. In order for a track to be successful, a fairly high contrast and unique region of the image must be located in the footage. This process is known as pattern selection.

The first step in pattern selection is to review the footage to be tracked several times. Watch for candidate patterns that are visible through the entire range of frames, where the contrast is high and the shape of the pattern does not change over time. The more unique the pattern, the more likely the track is to be successful.

In addition to locating high contrast, defined patterns, watch for the frames where the pattern moves the most. Identifying the maximum range of a pattern's motion will help to determine the correct size for the pattern search area.

It is not uncommon to have a scene that requires the use of several different patterns to generate a single path. This most often occurs because the pattern moves out of frame or is temporarily obscured by another scene element. Combining patterns into a single pattern is described later in the chapter.

Selecting the Pattern's Image Channels

When a pattern of pixels is selected, the Tracker automatically selects the color channel used for tracking the pattern based on an analysis of each channel for contrast, clarity, and reliability. The channels selected are highlighted in the bars to the right of the Pattern display window in the node controls.



Highlighted channel bars indicate which channel is selected for tracking

You can override the automatic channel selection by clicking the buttons beneath the bars for each channel to determine the channel used for tracking.

You can choose any one of the color channels, the luminance channels, or the alpha channel to track a pattern.

When choosing a channel, the goal is to choose the cleanest, highest contrast channel for use in the track. Channels that contain large amounts of grain or noise should be avoided. Bright objects against dark backgrounds often track best using the luminance channel.

Selecting Patterns for Stabilization

Selecting patterns for stabilization can be a tricky business. The location of the pattern, when it is selected, is used to determine precisely how the image will be stabilized. At least two patterns are required to correct for rotation; using three patterns will correct for scaling, and more will usually improve the quality of the solution.

Try not to select just any potentially valid pattern in the sequence, as some patterns will make the solution worse rather than better. To help with your selection, use the following guidelines when selecting patterns for stabilization.

- Locate patterns at the same relative depth in the image. Objects further in the background will move in greater amounts compared to objects in the foreground due to perspective distortion. This can confuse the stabilization calculations, which do not compensate for depth.
- Locate patterns that are fixed in position relative to each other. Patterns should not be capable of moving with reference to each other. The four corners of a sign would be excellent candidates, while the faces of two different people in the scene would be extremely poor choices for patterns.

Using the Pattern Flipbooks

Each pattern has a pair of thumbnail windows shown in the Inspector. The left window shows the selected pattern, while the right window is updated during the track to show the actual pattern that has been acquired for each frame.

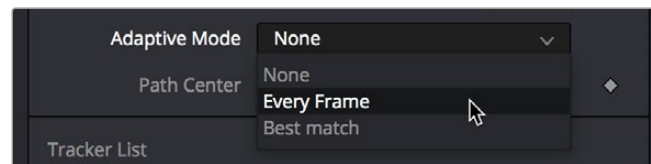


The Tracker Pattern Selection and Flipbook thumbnails

Each pattern that's stored is added to a Flipbook. Once the render is complete, you can play this Pattern Flipbook to help you evaluate the accuracy of the tracked path. If you notice any jumps in the frames, then you know something probably went wrong.

Using Adaptive Pattern Tracking

Even the most ideal pattern will usually undergo shifts in profile, lighting conditions, and other variables. These can adversely affect pattern recognition to the point that a pattern becomes unusable. The Tracker offers three modes of pattern acquisition during tracking that can help to correct these conditions. The modes can be set using the Adaptive Mode options in the Inspector.



The Adaptive Mode options

None

When the Adaptive mode is set to None, the pattern within the rectangle is acquired when the pattern is selected, and that becomes the only pattern used during the track.

Every Frame

When Every Frame is chosen, the pattern within the rectangle is acquired when the pattern is selected, and then reacquired at each frame. The pattern found at frame 1 is used in the search on frame 2, the pattern found on frame 2 is used to search frame 3, and so on. This method helps the Tracker adapt to changing conditions in the pattern.

Every Frame tracking is slower and can be prone to drifting from sub-pixel shifts in the pattern from frame to frame. Its use is therefore not recommended unless other methods fail.

Best Match Tracking

Best Match tracking works in much the same way as Every Frame tracking; however, it will not reacquire the pattern if the difference between the original pattern and the new one is too great. This helps to prevent cases where transient changes in the image cause the Tracker to become confused.

As a comparison between the two Adaptive modes, if a shadow passes over the tracker point, the Every Frame tracking mode may start tracking the shadow instead of the desired pattern. The Best Match mode would detect that the change from the previous frame's pattern was too extreme and would not grab a new pattern from that frame.

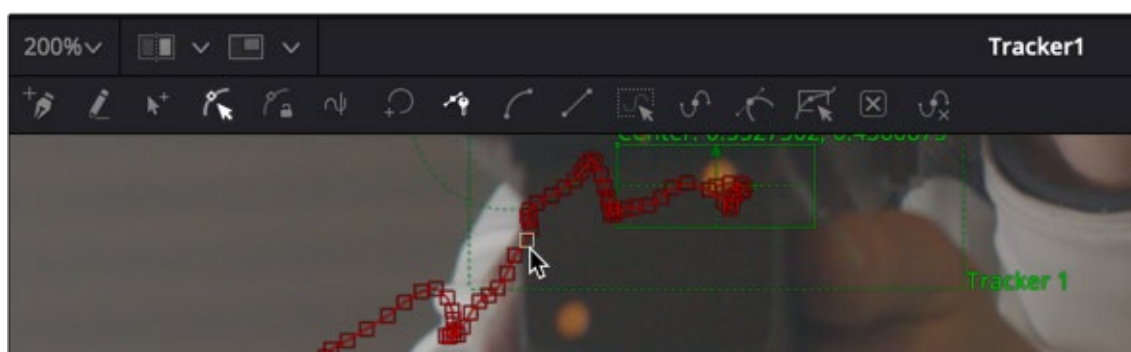
The Adaptive mode is applied to all active patterns while tracking. If you only want some patterns to use the Adaptive mode, disable all other patterns in the list before tracking.

Dealing with Obscured Patterns

Often, an otherwise ideal pattern can be temporarily obscured (occluded) or blocked from tracking—for example, when tracking a car that passes behind a telephone pole.

In these situations, you divide the render range up into two ranges, the range before the pattern is obscured and the range after the pattern becomes visible again. After tracking the two ranges individually, the Tracker will automatically interpolate between the end of the first range and the start of the second.

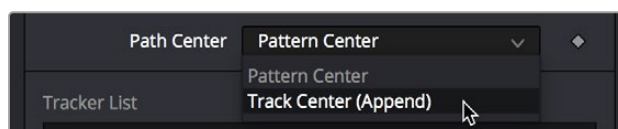
If you need to edit the resulting motion path to account for any non-linear motion that takes place between the two tracked ranges, you can select the track path to expose a Node toolbar with controls for adjusting the control points on this path. For example, you can choose Insert and Modify mode to insert points in the non-tracked range to compensate for any nonlinear motion in the tracked pattern.



Tools for modifying tracker paths in the Node toolbar of the viewer

Dealing with Patterns That Leave the Frame

There are two options when a tracker leaves the frame. If the pattern re-enters the frame, you can treat it like an obscured pattern. If the pattern does not re-enter the frame, or it is undesirable to hand track portions of the movement, you can use the Track Center (Append) mode to select a new pattern.



The Track Center (Append) mode pop-up menu

The Track Center (Append) mode selects a new pattern that will continue to add keyframes to the existing path. The offset between the old pattern and the new pattern is automatically calculated to create one continuous path.

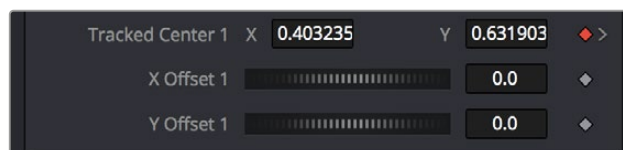
To use the Track Center (Append) mode, do the following:

- 1 When the pattern has become untrackable for some reason, stop analysis and move the playhead to the last frame that tracked successfully.
- 2 Choose Track Center (Append) from the Path Center pop-up menu in the Inspector.
- 3 Now, drag the Pattern selector to a new pattern that can be tracked from that point onward.
- 4 Restart tracking from the current frame.

When selecting a pattern to use in appending to an existing path, a pattern that is close to the old pattern and at the same apparent depth in the frame generates the best results. The further away the new pattern is, the more likely it is that the difference in perspective and axial rotation will reduce accuracy of the tracked result.

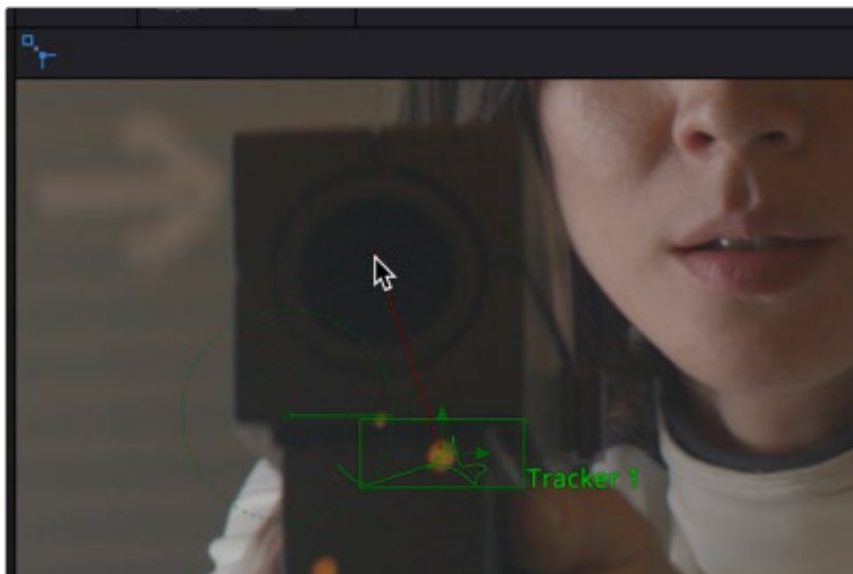
Setting Up Tracker Offsets

Often, it's impossible to track the thing you want to apply an effect to. For example, the only pattern available for an accurate track is a button on an actor's sleeve. However, the effect requires the person's hand to be glowing. To cause the glow's effect mask to be centered on the actor's hand, it's necessary to use the Tracker Offset control.



The Tracker Offset controls in the Inspector

The X and Y Offset controls allow for constant or animated positional offsets to be created relative to the actual Tracker's pattern center. The position of the offset in the viewer will be shown by a dashed line running from the pattern center to the offset position. You can also adjust the offset in the viewer using the Tracker Offset button. Clicking the button enables you to reposition the path while keeping the Tracker pattern in place.



The Tracker Offset tool in the Node toolbar of the viewer; a track of the orange dot is being offset to the center of the ray gun

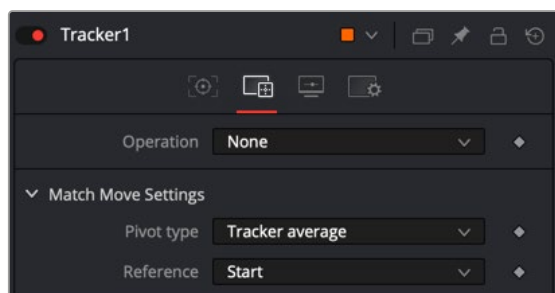
Once an offset for a pattern is set, you can connect other positional controls to the Tracker's Offset menu using the Connect To > Tracker: Offset Position option in the control's contextual menu. The path created during the track remains fixed to the center of the pattern.

Stabilizing with the Tracker Node

When a Tracker node is set to Match Move in the Operations tab, it is capable of a variety of functions. Applying the motion from the background clip to the foreground clip is the obvious functionality. However, the Match Move operation is also used for stabilizing footage to either completely remove motion from the scene or smooth existing motion.

Here are some common scenarios for stabilization that are handled when the Tracker is set to Match Move.

- A sequence that should be steady has vibrations or undesirable movement.
- A sequence that requires a smooth camera move suffers from jarring.



The Tracker Operation tab Match Move set to BG Only for stabilization

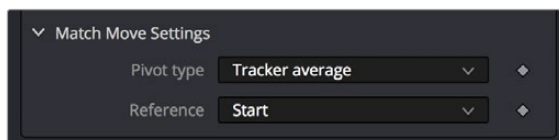
Stabilization Using the Tracker Match Move Mode

Stabilizing motion completely removes the appearance of motion from the image. The motion from frame to frame is calculated, and the contents of the frame are transformed to return the image to a reference position. This position can be either the start or end of the sequence or a manually selected frame from the sequence.

Stabilization can correct for position with as little as one pattern. Two or more patterns are required to correct for rotation or scaling within the image.

When the Operation menu is set to Match Move, choosing BG only from the Merge operation menu stabilizes the background (yellow input) clip. Only the controls that are applicable for stabilization operations will appear in the Operation tab.

Several of the stabilization controls are always available, collected under the Match Move Settings disclosure button. These controls are available at all times because the Steady and Unsteady positions of a tracker are always published. This makes them available for connection by other controls, even when the Tracker's operation is not set to match moving.



The Match Move settings

Merge

The Merge menu determines to which input connection the Tracking data is applied. When stabilizing an image to remove all motion, or smooth the motion, the Merge button must be set to BG Only.

Edges

The Edges menu determines whether the edges of an image that leave the visible frame are cropped, duplicated, or wrapped when the stabilization is applied. Wrapping edges is often desirable for some methods of match moving, although rarely when stabilizing the image for any other purpose.

For more information on the controls, see *Chapter 119, “Tracking Nodes,”* in the DaVinci Resolve Reference Manual or Chapter 59 in the Fusion Reference Manual.

Position/Rotation/Scaling

Use the Position, Rotation, and Scaling checkboxes to select what aspects of the motion are corrected.

Match Move Settings

Options for the Match Move settings include Pivot and Reference.

Pivot Type

The Pivot Type for the stabilization is used to calculate the axis of rotation and scaling calculations. This is usually the average of the combined pattern centers but may be changed to the position of a single tracker or a manually selected position.

Reference

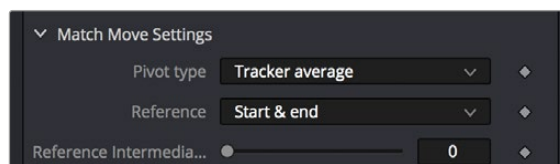
The Reference controls establish whether the image is stabilized to the first frame in the sequence, the last frame, or to a manually selected frame. Any deviation from this reference by the tracked patterns is transformed back to this ideal frame.

As a general rule, when tracking to remove all motion from a clip, set the Merge mode to BG Only, the Pivot Type to Tracker Average or Selected Tracker, and the Reference control to Start, End, or Select Time.

Smoothing Motion

When confronted with an image sequence with erratic or jerky camera motion, instead of trying to remove all movement from the shot, you often need to preserve the original camera movement while losing the erratic motion.

The Start & End reference option is designed for this technique. Instead of stabilizing to a reference frame, the tracked path is simplified. The position of each pattern is evaluated from the start of the path and the end of the path along with intervening points. The result is smooth motion that replaces the existing unsteady move.



The Reference Intermediate Points slider is displayed when Start & End is selected to enable the smoothing of motion

To preserve some of the curvature of the original camera motion, you can increase the value of the Reference Intermediate Points slider that appears when the Start & End reference mode is selected.

When tracking to create smooth camera motion, ensure that the Start & End reference mode is enabled and set the Merge mode to BG Only. It is recommended to leave the Pivot Type control set to Tracker Average.

Using the Tracker Node for Match Moving

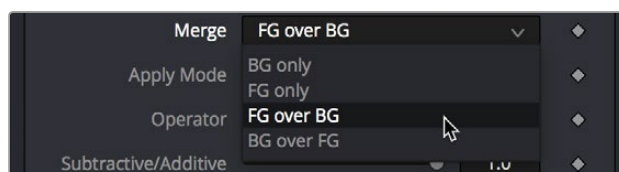
A simple match moving example is shown at the beginning of this chapter, but this section presents additional details that you may not have been aware of. Examples of match moving include:

- A static CG element must be believably added to a moving sequence.
- Two sequences with different motions must be composited together.

Some clips may need to be stabilized so that an element from another source can be added to the shot. After the element or effect has been composited, the stabilization should be removed to make the shot look natural again.

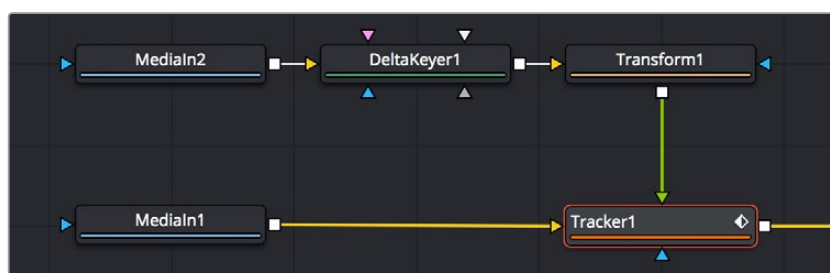
Simple Match Moving

Match moving essentially applies the movement from the tracked clip to another clip. There are two ways to perform match moving. One method involves connecting other nodes, such as Transform or Merge, to a Tracker's outputs. The other method is to stabilize an image by trying to remove all motion, but instead of setting the Merge menu to BG Only, set it to FG Over BG, FG Only, or in rare occasions, BG Over FG.



Set the Merge menu to BG Only, FG Over BG, or BG Over FG

When using this Merge menu, you connect a foreground image to the Tracker node's input connection in the Node Editor.



Connect a foreground image to the Tracker's foreground input

Enabling the FG Only mode will apply the motion from the background to the foreground, and the Tracker will only output the modified FG image. This result can later be merged over the original, allowing further modifications of the foreground to be applied using other nodes before merging the result over the background clip.

Corner Positioning Operations

The Corner Positioning operation maps the four corners of a foreground image to four patterns within the Tracker. This operation, or technique, is most commonly used for sign replacements.

The Corner Positioning operation of the Tracker requires the presence of a minimum of four patterns. If this operation mode is selected and there are not four patterns set up in the Tracker already, additional patterns will automatically be added to bring the total up to four.

When this mode is enabled, a set of drop-down boxes will appear to select which tracker relates to each corner of the rectangle. It has no effect when the Merge control option is set to BG Only.

Perspective Positioning Operations

The Perspective Positioning operation is used to remove perspective from a foreground image or apply the perspective from one sequence to another. This can be useful when you need to paint out an area that is distorted by perspective. Removing the perspective flattens the images for painting, and then another tracker adds the perspective back.

The Perspective Positioning operation of the Tracker requires the presence of a minimum of four patterns. If this operation mode is selected and there are not four patterns set up in the Tracker already, additional patterns will automatically be added to bring the total up to four.

When this mode is enabled, a set of drop-down boxes will appear to select which tracker relates to each corner of the rectangle. It has no effect when the Merge control option is set to BG Only.

Connecting to Trackers' Operations

One of the most common applications for a tracked pattern is using the tracked position or path to drive the position of another node's parameters. For example, tracking an eye in order to color correct the eye to blue using an effect mask. You start off by tracking the eye, and then create a color corrector with the desired settings. You create a mask in the shape of the eye and connect the Tracker's position to the Center of the mask.

In addition to the path (called Offset Position), each pattern in a tracker publishes four other values for use as connections that are available to other nodes in the Node Editor.

You connect a node's position parameters to a tracker by selecting the connection type from the controls contextual menu (for example, Transform 1: Center > Connect To > Tracker 1 > Offset Position).

There are five connection types automatically published by the tracker to connect to a position parameter in another node.

Steady Position

Steady Position can be used to stabilize footage in both X and/or Y to remove camera shake and other unwanted movement. The connection inverts the output of the tracked pattern's motion. When you connect a Center parameter to the Steady Position of the Tracker, it will be placed at 0.5/0.5 (the center of the screen) by default at frame 1. You can change this using the Reference mode in the Tracker's Operation tab.

Steady Angle

The Steady Angle mode can be used to stabilize footage in both X and/or Y to remove camera shake and other unwanted movement. When you connect a control, for example the Angle of a Transform, to the Steady Angle of the Tracker, it will be placed at 0 degrees by default at frame 1. This can be

changed by means of the Reference mode in the Tracker's Operation tab. From there on, the resulting motion of the Steady Angle mode will rotate into the opposite direction of the original motion.

So if the angle at frame 10 is 15 degrees, the result of the Steady Angle will be -15 degrees.

To use Steady Angle, you need at least two tracked patterns in your tracker. With just one point, you can only apply (Un)Steady Position.

Offset Position

An Offset Position is essentially the path generated by the tracker. It is the one you select when you want an object to follow the path. It is available for each single tracker in the Tracker node and refers to that single tracker only. When you connect the Center X and Y parameters to the offset position of the Tracker, the node's center will follow exactly the path of that tracker. Connecting to single trackers is always useful when you want to match elements with object motion in your footage. For example, you could track a hand of your actor and attach a ball to the Tracker's offset position, so that the ball follows the exact motion of the hand. Or you could track an element that needs rotoscoping and attach the mask's center to the Tracker's offset position.

Unsteady Position

After using the Steady Position, the Unsteady Position is used to reintroduce the original movement on an image after an effect or new layer has been added. The resulting motion from Unsteady Position is basically an offset in the same direction as the original motion.

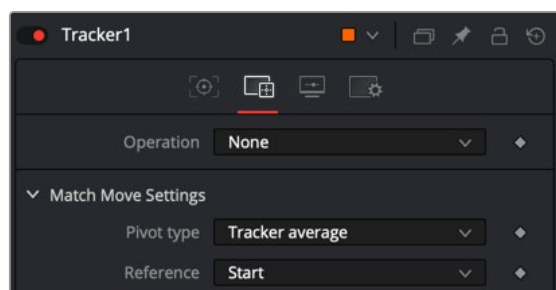
Steady Size

The Steady Size connection outputs the inverse of the tracked pattern's scale. When you connect a parameter, for example the Size of a Transform, to the Steady Size of the Tracker, it will be placed with a Size of 1 (i.e., the original size) by default at frame 1. This can be changed by means of the Reference mode in the Tracker's Operation tab. The resulting size of the Steady Size mode will then counteract the size changes of the original motion. So if the actual size at frame 10 is 1.15, the result of the Steady Size will be $1 - (1.15 - 1) = 0.85$.

To use Steady Size, you need at least two tracked patterns in your tracker. With just one point, you can only apply (Un)Steady Position.

Using the Outputs of a Tracker

The tracker outputs described above are published by each tracker pattern created in the Tracker node. Each Tracker node itself also publishes a Steady Position, Angle, Size and an Unsteady Position. The values of these Tracker node outputs are calculated using all the patterns in that tracker, as configured by the Match Move Settings controls in the Tracker's Operation tab.



The Match Move settings determine the reference frame for stabilization and the point around which stabilization pivots

Rather than using the Tracker node to perform the Merge operation, an alternative and common way to use these published outputs is to create a match move by connecting the outputs to multiple nodes. A tracker is used to track a pattern, and then that data can be connected to multiple other nodes using the Connect To submenu.

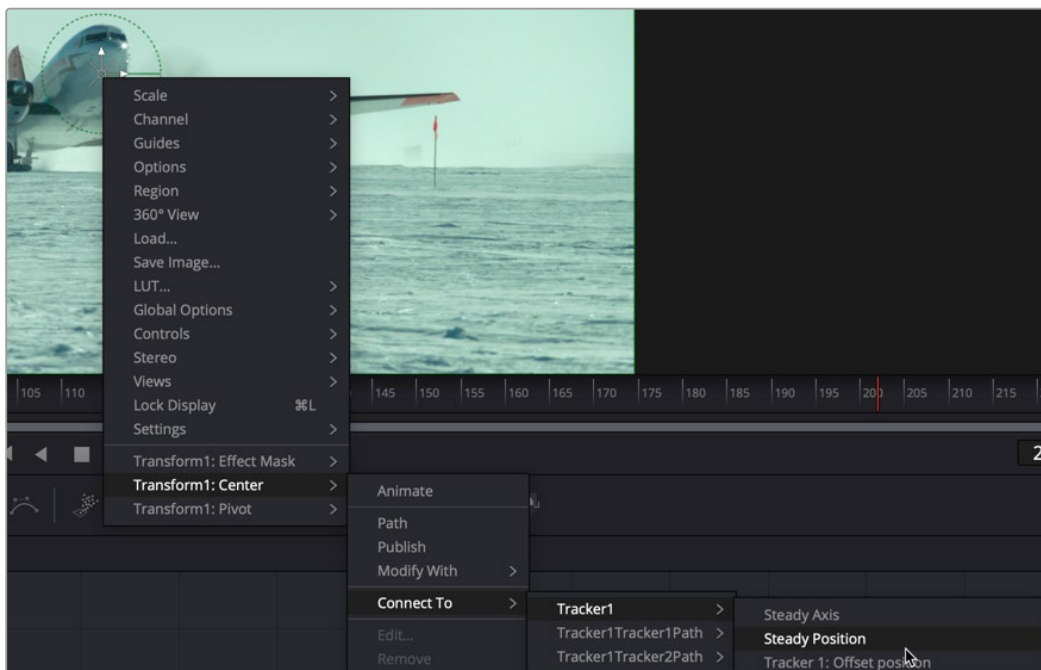
As an example, to use the Connect To menu to perform a match move, do the following:

- 1 Track the background clip using at least two tracking patterns in the tracker.
- 2 In a different branch, add a Transform node to the background clip.



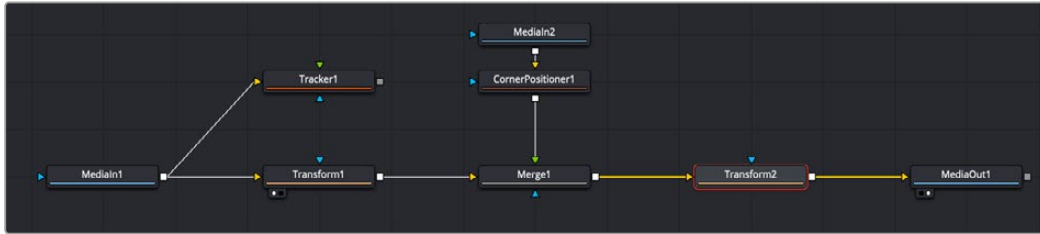
The background branched to go to a Tracker and a Transform

- 3 Right-click over the Transform's Center and choose Connect to > Tracker1 > Steady Position.



The tracker publishes its output for other nodes to connect to, as done here to stabilize the clip

- 4 Connect the foreground to a corner-positioned node, so you can position the corners of the foreground appropriately over the background.
- 5 Add another Transform node to the Node Editor after the Merge.



A second Transform after the Merge is used to add back in the original motion with Unsteady Position

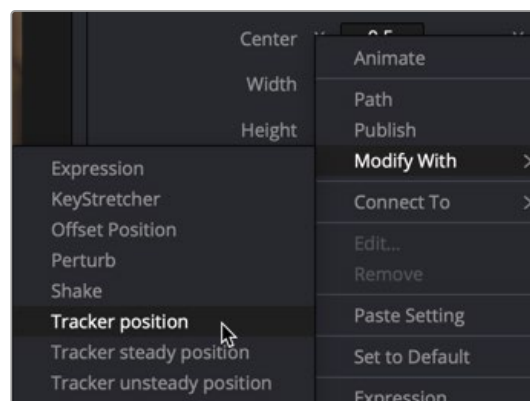
- 6 Connect the new Transform's Center to the Tracker's Unsteady Position. The image will be restored to its original state with the additional effect included.

To better understand how this works, imagine a pattern that is selected at frame 1, at position 0.5, 0.5. The pattern does not move on frame 2, so its position is still 0.5, 0.5. On the third frame, it moves 10 percent of the image's width to the right. Now its position is 0.6, 0.5.

If a transform center is connected to the Steady Position output of the Tracker, the Transform node's center is 0.5, 0.5 on the first and second frames because there has been no change. On frame 3, the center moves to 0.4, 0.5. This is the inverse of the horizontal motion that was tracked in the pattern, moving the image slightly to the right by 10 percent of the image width to counteract the movement and return the pattern of pixels back to where they were found.

Using the Tracker as a Modifier

Another technique for adding a tracker directly to a control is to add it as a modifier. Choosing the Tracker from the Modify With contextual menu does not use a Tracker node; it adds a modifier in the Inspector with a set of parameters almost identical to those found in the Tracker node itself. The benefit here is that the object that you want to follow the tracked path is automatically connected to the tracker modifier when you apply it.



Applying the tracker as a modifier

The differences between a Tracker modifier and a Tracker node are as follows:

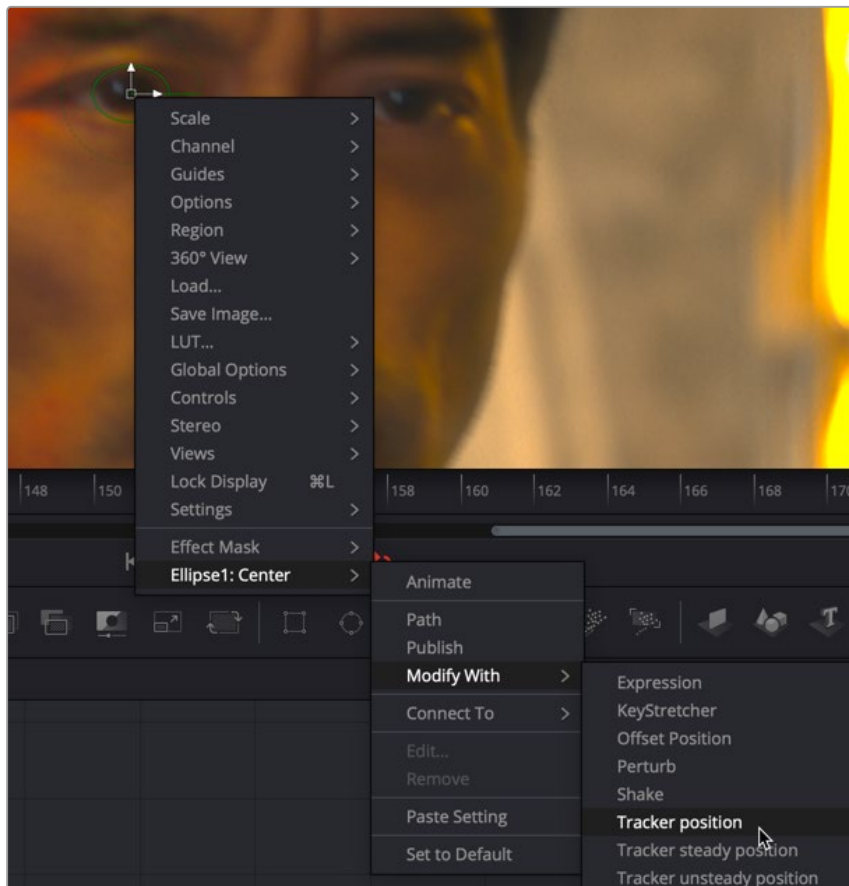
- The Tracker modifier can only track a single pattern.
- A source image must be set for the Tracker modifier.

The Tracker modifier can only output a single value and cannot be used for complex stabilization procedures, but it is a nice quick way to apply a tracker to a point that you need to follow.

As an example, to apply the Tracker as a modifier, do the following:

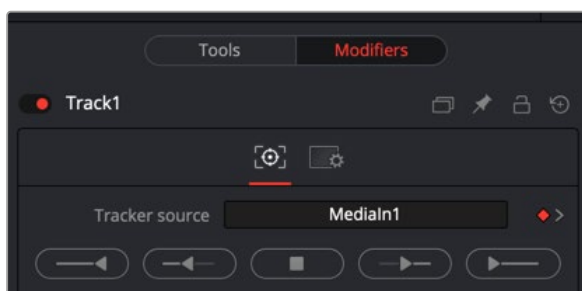
Imagine that you needed to track an actor's eyes so that an unearthly, alien glow could be applied to the eyes.

- 1 Add an ellipse mask node to cover an actor's eye.
- 2 In the Inspector, right-click on the mask's Center parameter and from the contextual menu choose Ellipse1 Center > Modify With > Tracker Position.



Right-click over any Center Coordinate Control and choose Modify With > Tracker Position to add a Tracker modifier

- 3 Click the Modifiers tab in the Inspector and drag the MediaIn1 node that you want to track into the Tracker Source field.



The Modifiers tab includes the tracking controls and a Tracker Source field to indicate which node to use for tracking

- 4 Click the Track Forward button to begin tracking the person's eye.
- 5 Insert a Soft Glow node directly after the MediaIn and connect the Ellipse Mask to the white Glow Mask input.



A Tracker modifier applied to the Ellipse to create a green glow on an actor's pupil

You can set a different source image for the Tracker modifier by typing in the name of the node or dragging and dropping the node from the Node Editor into the Tracker Source field control. If you have a node (let's call it node#1) connected to the node that contains the modifier (let's call it node#2), the source image for the Tracker modifier will automatically be the node #1

For more information on the Tracking parameters, see *Chapter 119, "Tracking Nodes,"* in the DaVinci Resolve Reference Manual or Chapter 59 in the Fusion Reference Manual.

Match Moving Text Example

This example takes you through a complete motion tracking task, and shows how you can create a very simple match-moving effect using the Tracker node, which is the Swiss army knife of trackers in Fusion.

Adding a Layer to Match Move

In the example composition, we have a Text1 node that's creating a "Switzerland" title that's composited over a drone shot flying over and around a mountain bridge. With the Text1 node selected, the onscreen controls that let you position the text it's generating are visible in the viewer, and the text is positioned where we'd like it to start. Note that, with the Text node selected, even the part of the text that's offscreen can still be seen as an outline showing us where it is.

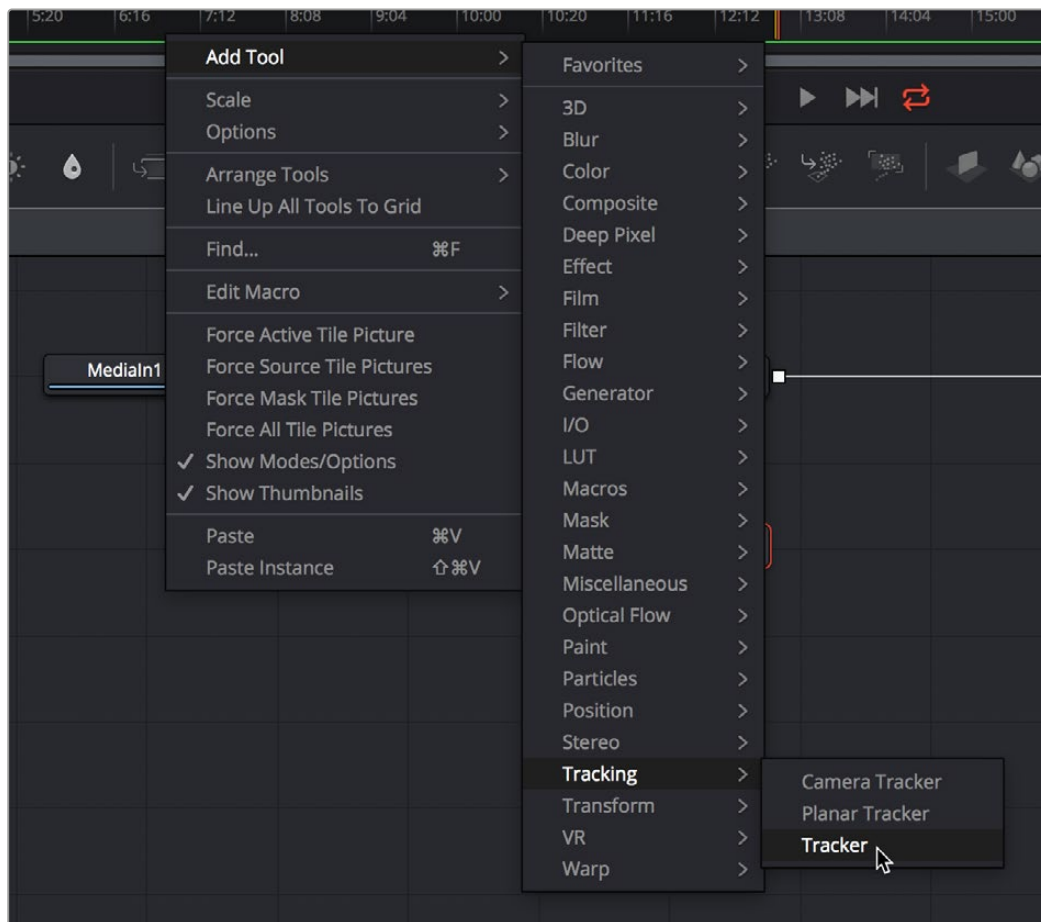


Some text superimposed against a background, ready to track

Our goal for this composition is to motion track the background image so that the text moves along with the scene as the camera flies along.

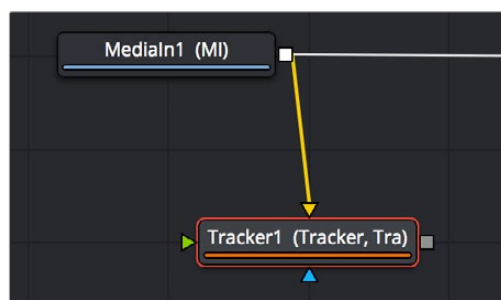
Setting Up Motion Tracking

To set up for the motion track, we'll begin by creating a disconnected Tracker node, using another method other than those seen previously. Right-click anywhere in the background of the Node Editor (preferably where you want the new node to appear), and choose Add Tool > Tracking > Tracker from the contextual menu to create a new Tracker1 node underneath the MediaIn node (or Loader node if you are using Fusion Studio).



Creating a new node using the Node Editor contextual menu

Next, we'll drag a connection from the Medialn1 node to the Tracker1 node to automatically connect the source clip to the Tracker1 background input. This branches the output from the Medialn1 node to the Tracker node so that the Tracker1 node processes the image separately from the rest of the node tree. This is not required, but it's a nice organizational way to see that the Tracker node is doing an analysis that must be referred to in a way other than a "physical" connection.



Branching a Tracker node to use to analyze an image

A Simple Tracking Workflow

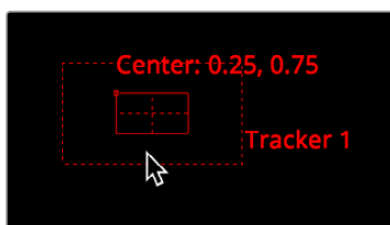
The Tracker node is the simplest tracking operation the Fusion page has, and while there are several ways of using it, an extremely common workflow is to use the Tracker node controls to analyze the motion of a subject in the frame with motion you want to follow, and then use the resulting motion path data by “connecting” it to the Center parameter of another node that’s capable of transforming the image you want to match move.

Positioning the Tracker Onscreen Control

When the Tracker node is selected, a single green box appears in the viewer, which is the default onscreen control for the first default tracker that node contains (seen in the Tracker List of the Inspector controls). Keep in mind that you only see onscreen controls for nodes that are selected, so if you don’t see the onscreen tracker controls, you know you need to select the tracker you want to work with. Loading the tracker you want to work with into the viewer is also the safest way to make sure you’re positioning the controls correctly relative to the actual image that you’re tracking.

If you position your pointer over this box, the entire onscreen control for that tracker appears, and if you click the onscreen control to select that tracker, it turns red. As with so many other tracker interfaces you’ve likely used, this consists of two boxes with various handles for moving and resizing them:

- The inner box is the “pattern box,” which identifies the “pattern” in the image you’re tracking and want to follow the motion of. The pattern box has a tiny handle at its upper-left corner that you use to drag the box to overlap whatever you want to track. You can also resize this box by dragging any corner, or you can squish or stretch the box by dragging any edge to make the box better fit the size of the pattern you’re trying to track. The center position of the tracker is indicated via X and Y coordinates.
- The outer box is the “search box,” which identifies how much of the image the tracker needs to analyze to follow the motion of the pattern. If you have a slow-moving image, then the default search box size is probably fine. However, if you have a fast-moving image, you may need to resize the search box (using the same kind of corner and side handles) to search a larger area, at the expense of a longer analysis. The name of that tracker is shown at the bottom right of the search box.



The onscreen controls of a selected tracker seen in isolation

It’s worth saying a second time that the handle for moving a tracker’s onscreen control is a tiny dot at the upper-left corner of the inner pattern box. It’s really easy to miss if you’re new to Fusion. You must click on this dot to drag the tracker around.



The handle for dragging the tracker boxes to move them around

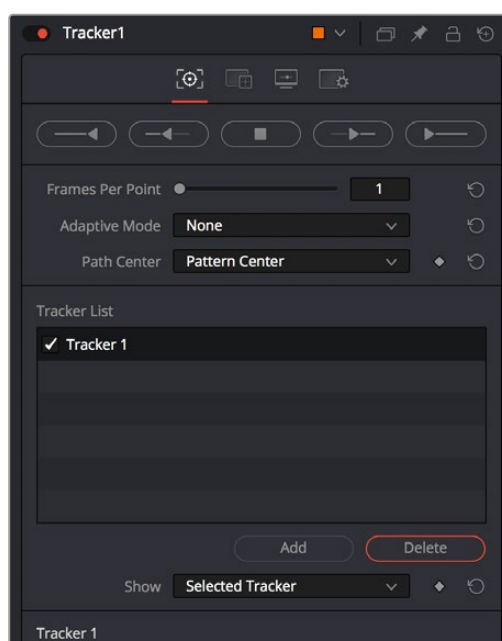
In this example, we'll drag the onscreen control so the pattern box overlaps a section of the bridge right over the leftmost support. As we drag the onscreen control, we see a zoomed-in representation of the part of the image we're dragging over to help us position the tracker with greater precision. For this example, the default sizes of the pattern and search box are fine as is.



The zoomed-in preview that helps you position the pattern box as you drag it

Using the Tracker's Inspector Controls to Perform the Analysis

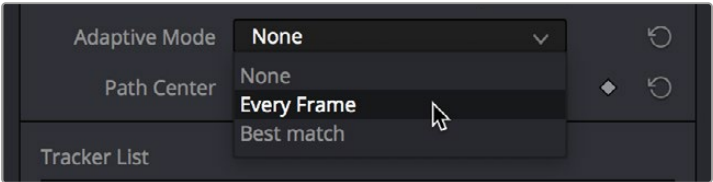
At this point, let's look at the Tracker node's controls in the Inspector. There are a lot of controls, but for this simple example we only care about the main Tracker panel, with the tracking analysis buttons at the top, the tracking options below those, and the Tracker List underneath those. The Tracker List also has buttons for adding and deleting trackers; you have the option of adding multiple trackers that can be analyzed all at once for different workflows, but we don't need that for now.



Tracker Inspector controls, with the tracking analysis buttons at top, the tracker options in the middle, and the Tracker List below

Additional controls over each tracker and the image channels being analyzed appear at the bottom, along with offset controls for each tracker, but we don't need those now (at least not yet).

Again, this track is so simple that we don't need to change the default behaviors that much, but because the drone is flying in a circular pattern, the shape of the pattern area is changing as the clip plays. Fortunately, we can choose Every Frame from the Adaptive Mode menu to instruct the tracker to update the pattern being matched at every frame of the analysis, to account for this.



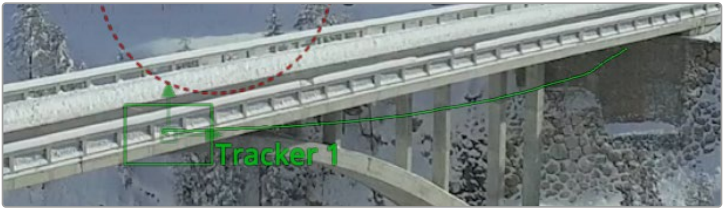
Changing the Adaptive Mode of the Tracker node to Every Frame to account for the camera's shift of perspective

Now, we just need to use the tracker analysis buttons at the top to begin the analysis. These buttons work like transport controls, letting you start and stop analysis as necessary to deal with problem tracks in various ways. Keep in mind that the first and last buttons, Track from Last Frame and Track from First Frame, always begin a track at the last or first frame of the composition, regardless of the playhead's current position, so make sure you've placed your tracker onscreen controls appropriately at the last or first frame.



The analysis buttons, left to right: Track from Last Frame, Track Backward, Stop Tracking, Track Forward, Track from First Frame

For now, clicking the Track from Beginning button will analyze the entire range of this clip, from the first frame to the last. A dialog lets you know when the analysis is completed, and clicking the OK button dismisses it so you can see the nice clean motion path that results.



The analyzed motion path resulting from tracking a section of the bridge as the camera flies past

Viewing Motion Track Data in the Spline Editor

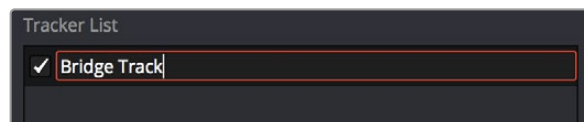
This is not a necessary part of the tracking workflow, but if you have an otherwise nice track with a few bumps in it, you can view the motion tracking data in the Spline Editor by viewing that tracker's Displacement parameter curve. This curve is editable, so you can massage your tracking data in a variety of ways, if necessary.



Viewing motion tracking analysis data in the Spline Editor

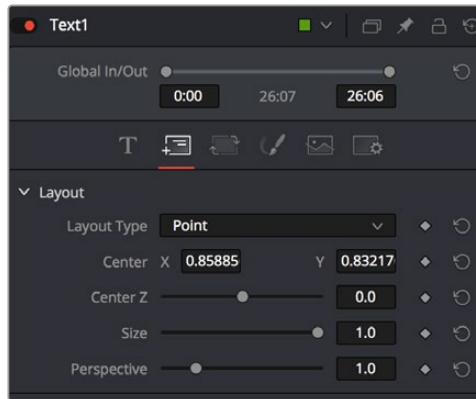
Connecting Motion Track Data to Match Move

Now that we have a successful analysis, it's time to use it to create the Match Move effect. To make this process easier, we'll double-click the tracker's name in the Tracker List of the Inspector, and enter a new name that's easier to keep track of (heh). Adding your own names make that tracker easier to find in subsequent contextual menus and lets you keep track of which trackers are following which subjects as you work on increasingly complex compositions.



Renaming a tracker to make it easier to find

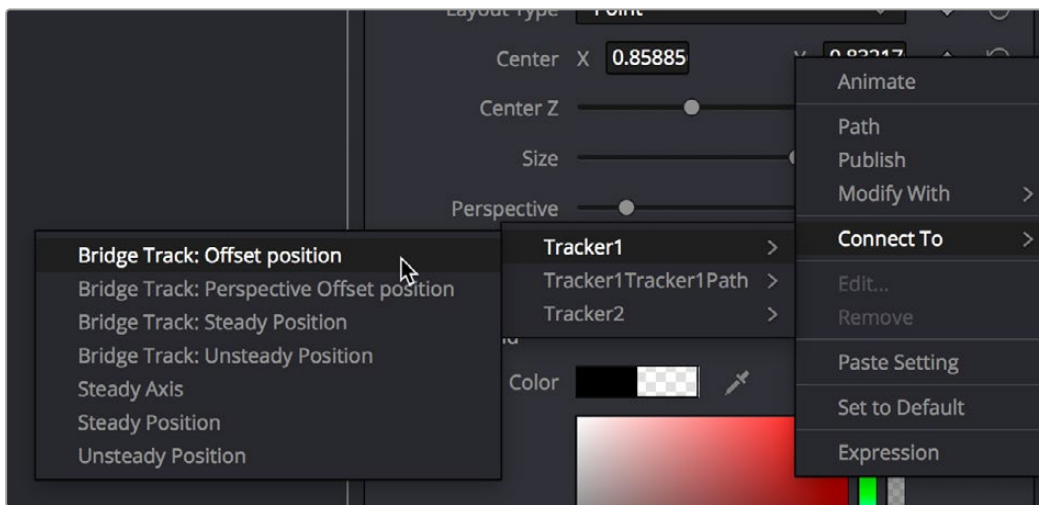
Now it's time to connect the track we've just made to the text in order to start it in motion. After loading the Merge1 node into the viewer to see the text in context with the overall composite we're creating, we'll select the Text1 node to open its parameters in the Inspector, and click the Layout panel icon (second button from the left) to expose the Layout controls, which are the text-specific transform controls used to position the text object in the frame. These are the controls that are manipulated when you use the Text node onscreen controls for repositioning or rotating text.



The Layout controls for a Text node, in the Layout panel

The Center X and Y parameters, while individually adjustable, also function as a single target for purposes of connecting to tracking to quickly set up match moving animation. You set this up via the contextual menu that appears when you right-click any parameter in the Inspector, which contains a variety of commands for adding keyframing, modifiers, expressions, and other automated methods of animation including connecting to motion tracking.

If we right-click anywhere on the line of controls for Center X and Y, we can choose **Connect To > Tracker1 > Bridge Track: Offset position** from the contextual menu, which connects this parameter to the tracking data we analyzed earlier.



Connecting the Center X and Y parameter to the Bridge Track: Offset position motion path we analyzed

Immediately, the text moves so that the center position coincides with the center of the tracked motion path at that frame. This lets us know the center of the text is being match moved to the motion track path.



The text now aligns with the motion track coordinate

Offsetting the Position of a Match Moved Image

In fact, we want to offset the match-moved text, so it's higher up in the frame. To do this, we select the Tracker1 node again and use the Y Offset 1 dial control to move the text up, since now any changes we make to the Bridge Track dataset now apply to the center of the text that's connected to it.



Using the X and Y Offset controls in the Tracker1 node to offset the text layer's position from the tracked motion path

The offset we create is shown as a dotted red line that lets us see the actual offset being created by the X and Y Offset controls. In fact, this is why we connected to the Bridge Track: Offset position option earlier.



The text offset from the tracked motion path; the offset can be seen as a dotted red line in the viewer

Now, if we play through this clip, we can see the text moving along with the bridge.



Two frames of the text being match moved to follow the bridge in the shot